

Louay Bannani

Data Science & AI/ML Engineer

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SUMMARY

AI & Data Science engineer who builds end-to-end intelligent systems, from data pipelines and forecasting models to LLM-powered multi-agent assistants deployed on real business channels. Delivered PrediBot, a multi-agent restaurant-operations platform (six MCP sub-agents, 21 n8n workflows, Prophet forecasting service, WhatsApp integration), and a CNN-based invoice OCR pipeline with financial anomaly detection. Strong in Python, deep learning, RAG, time series and REST API integration, with full-stack experience in Laravel, MySQL and Vue.js.

EXPERIENCE

Data Science & AI Engineer Intern

Tunis, Tunisia

PrendsTaPart

February 2026 – August 2026 (6 months)

- Architected and delivered **PrediBot**, a multi-agent AI system integrated with the FoodEatUp platform, enabling natural-language access to inventory, HACCP, suppliers, HR, and production management.
- Built and orchestrated **six MCP-based sub-agents** and **21 n8n workflows** connecting LLMs, REST APIs, authentication, and automated business processes.
- Developed a **Python/Flask forecasting microservice** with Prophet for recipe-level production forecasting, covering data preprocessing, per-recipe models, fallback strategies, and API integration (forecast error: **9%** MAPE).
- Deployed conversational access through **OpenAI Agent Builder, ChatKit, and Twilio WhatsApp**, with proactive operational alerts for kitchen and stock teams.

Data Science & AI Engineer Intern

Tunis, Tunisia

PrendsTaPart

July 2025 – August 2025 (2 months)

- Led the design and first prototype of **PrediBot**, an AI agent for restaurant management combining sales forecasting, inventory optimization, and HACCP anomaly detection.
- Automated data ingestion and processing pipelines with **n8n**, storing curated data in **PostgreSQL**.
- Trained predictive models for sales, stock levels, and temperature anomalies, and exposed results through REST APIs for real-time alerts.

Data Science & AI Engineer Intern

Tunis, Tunisia

Proxym

June 2025 – July 2025 (1 month)

- Developed a **CNN-based OCR system** extracting key financial fields from invoices (dates, totals, VAT, line items) into structured data (field-level accuracy: **93%**).
- Designed a **document layout analysis** workflow with deep learning and OpenCV to segment invoice components across heterogeneous templates.
- Implemented **financial anomaly detection** rules flagging inconsistent totals, incorrect VAT calculations, and suspicious invoice structures.
- Built an end-to-end pipeline (preprocessing, extraction, validation, standardized **JSON export**) ready for integration with accounting systems.

EDUCATION

École Pluridisciplinaire Internationale (EPI)

Sousse, Tunisia

Engineering Degree in Artificial Intelligence and Data Science

September 2021 – August 2026

PROJECTS

RAG-Based HR Assistant | Python, LangChain, OpenAI API, ChromaDB (2025)

- Built a Retrieval-Augmented Generation system answering HR questions from internal documents, with ingestion, chunking, and embedding-based semantic search.
- Constrained the LLM to retrieved context to produce grounded answers and reduce hallucinations.

Speech Emotion Recognition (CNN) | Python, TensorFlow, Librosa, Scikit-learn (2025)

- Built a preprocessing pipeline over four datasets (RAVDESS, CREMA-D, TESS, SAVEE): slicing, normalization, and MFCC/spectrogram feature extraction.
- Trained CNN classifiers for emotion recognition, improving results through hyperparameter tuning and data augmentation (accuracy: **91%**).

Hybrid Movie Recommendation System | Python, Scikit-learn, NLTK, Surprise (2024)

- Combined collaborative filtering (Surprise) with content-based filtering (NLP on metadata, cosine similarity) into a blended hybrid recommender .

Time Series Forecasting | Python, ARIMA, SARIMA, Prophet (2023)

- Applied decomposition and stationarity testing, then benchmarked ARIMA, SARIMA, and Prophet using RMSE and MAE to select the best model.

TECHNICAL SKILLS

Languages: Python, SQL, Java, PHP, JavaScript

Machine Learning & Deep Learning: TensorFlow, PyTorch, Keras, Scikit-learn, XGBoost, Prophet, OpenCV, Pandas, NumPy

LLMs & Agents: LangChain, RAG, ChromaDB, MCP, OpenAI Agent Builder, ChatKit, Prompt Engineering

Data Engineering & APIs: n8n, FastAPI, Flask, REST APIs, PostgreSQL, MySQL

Web Development: Laravel, Vue.js, MySQL

MLOps & Deployment: Docker, MLflow, Git, GitHub Actions (CI/CD)

Tools: Jupyter Notebook, VS Code, PyCharm, Postman, Twilio

CERTIFICATES

Microsoft Azure Fundamentals (AZ-900)

NVIDIA: Fundamentals of Deep Learning

NVIDIA: Computer Vision for Industrial Inspection

NVIDIA: Applications of AI for Anomaly Detection

NVIDIA: Fundamentals of Accelerated Data Science

Data Science Bootcamp

Cisco CCNA 1 & 2: Networks, Switching, Routing, and Wireless Essentials

SOFT SKILLS & LANGUAGES

Soft Skills: Communication & Presentation, Team Leadership, Project Management, Problem-Solving, Self-Learning

Languages: Arabic (Native), English (Professional), French (Intermediate)

ASSOCIATIVE LIFE

- Active Member – EPI Junior Entrepreneur, Project Division (2024–2025)
- Former Member – IEEE EPI Student Branch, Computer Society Chapter (2024)